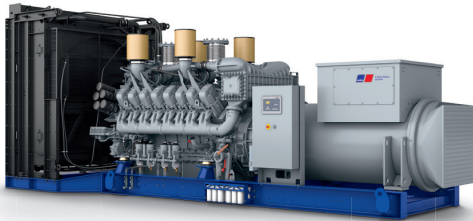




Diesel Generator Set

mtu 16V4000 DS2250

380V – 11 kV/50 Hz/prime power for stationary emergency/
NEA (ORDE) optimized/16V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 2040 kVA - 2160 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	300
Model		16V4000G14F	Engine jacket water capacity: l	175
Type		4-cycle	Intercooler coolant capacity: l	50
Arrangement		16V	Combustion air requirements	
Displacement: l		76.3	Combustion air volume: m³/s	2.4
Bore: mm		170	Max. air intake restriction: mbar	50
Stroke: mm		210	Cooling/radiator system	
Compression ratio		16.4	Coolant flow rate (HT circuit): m³/hr	68.5
Rated speed: rpm		1500	Coolant flow rate (LT circuit): m³/hr	30
Engine governor		ECU 9	Heat rejection to coolant: kW	610
Max power: kWm		1798	Heat radiated to charge air cooling: kW	370
Air cleaner		dry	Heat radiated to ambient: kW	90
Fuel system			Fan power for electr. radiator (40°C): kW	70
Maximum fuel lift: m		5	Exhaust system	
Total fuel flow: l/min		20	Exhaust gas temp. (after turbocharger): °C	475
Fuel consumption ²⁾			Exhaust gas volume: m³/s	6.2
At 100% of power rating:	l/hr	g/kwh	Maximum allowable back pressure: mbar	85
At 75% of power rating:	433.3	200	Minimum allowable back pressure: mbar	30
At 50% of power rating:	331.4	204		
	229.6	212		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer standard)	380 V	1728	2160	3282	1656	2070	3145
	400 V	1728	2160	3118	1656	2070	2988
	415 V	1728	2160	3005	1656	2070	2880
Marathon 744RSL7092 (Low voltage Marathon)	380 V	1704	2130	3236	1632	2040	3099
	400 V	1704	2130	3074	1632	2040	2944
	415 V	1696	2120	2949	1632	2040	2838
Marathon 1020FDL7093 (Low voltage Marathon oversized)	380 V	1704	2130	3236	1632	2040	3099
	400 V	1704	2130	3074	1632	2040	2944
	415 V	1696	2120	2949	1632	2040	2838
Marathon 1020FDH7097 (Medium volt. marathon)	11 kV	1712	2140	112	1632	2040	107
Leroy Somer LSA53.2 XL9 (Medium volt. Leroy Somer)	11 kV	1720	2150	113	1656	2070	109

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.